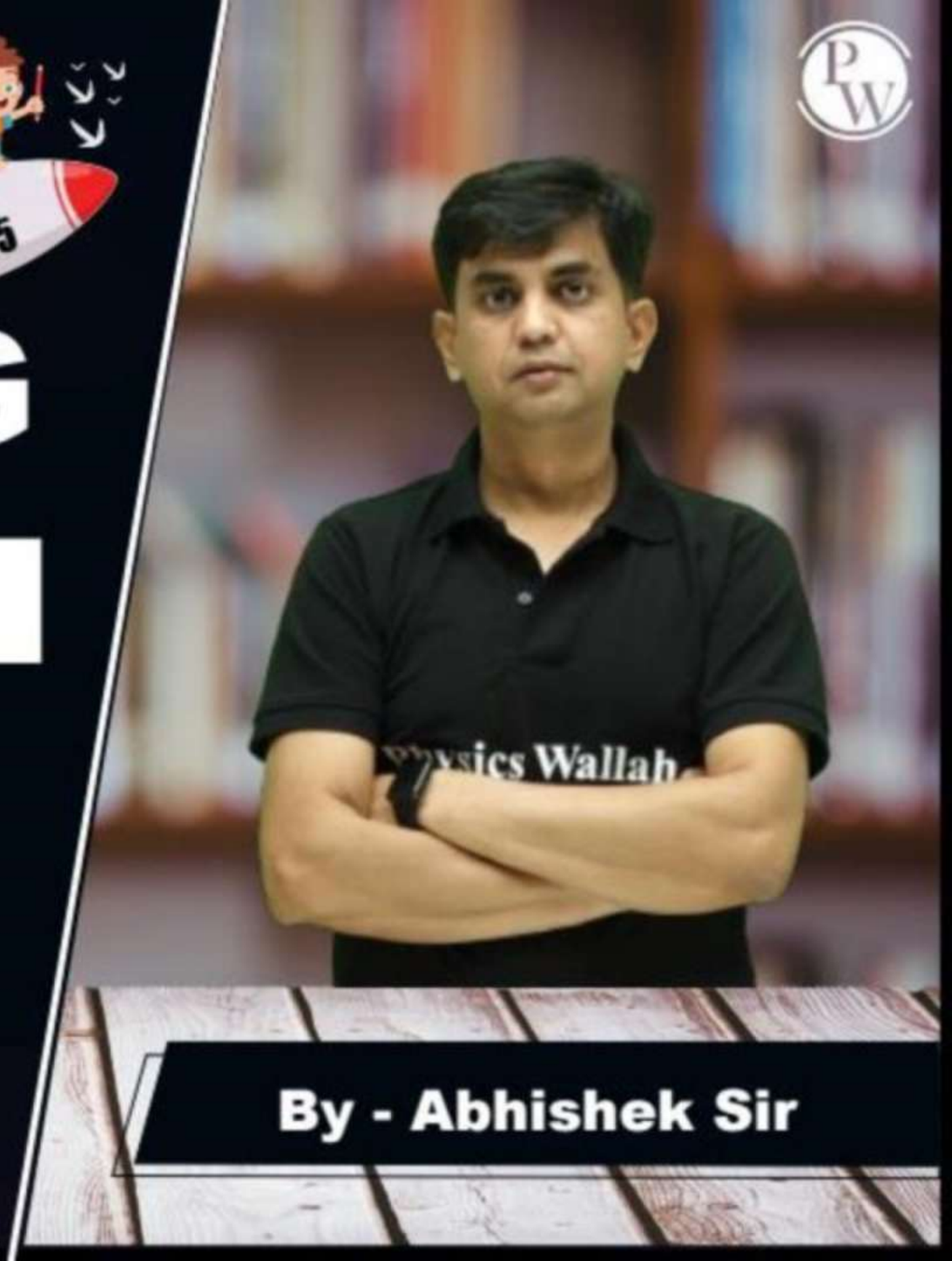
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 08



By - Abhishek Sir

Recap of Previous Lecture

Topic UDP Socket

Topic TCP Flow Control

Topics to be Covered

Topic TCP Congestion Control

Topic TCP Timers

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT?

I. A connected UDP socket can be used to communicate with multiple peers simultaneously. FALSE

II. A process can successfully call connect function again for an already connected UDP socket. TRUE

[GATE-2017, Set-2]

(A) I only

(B) II only

(C) Both I and II

(D) Neither I nor II

Ans: B

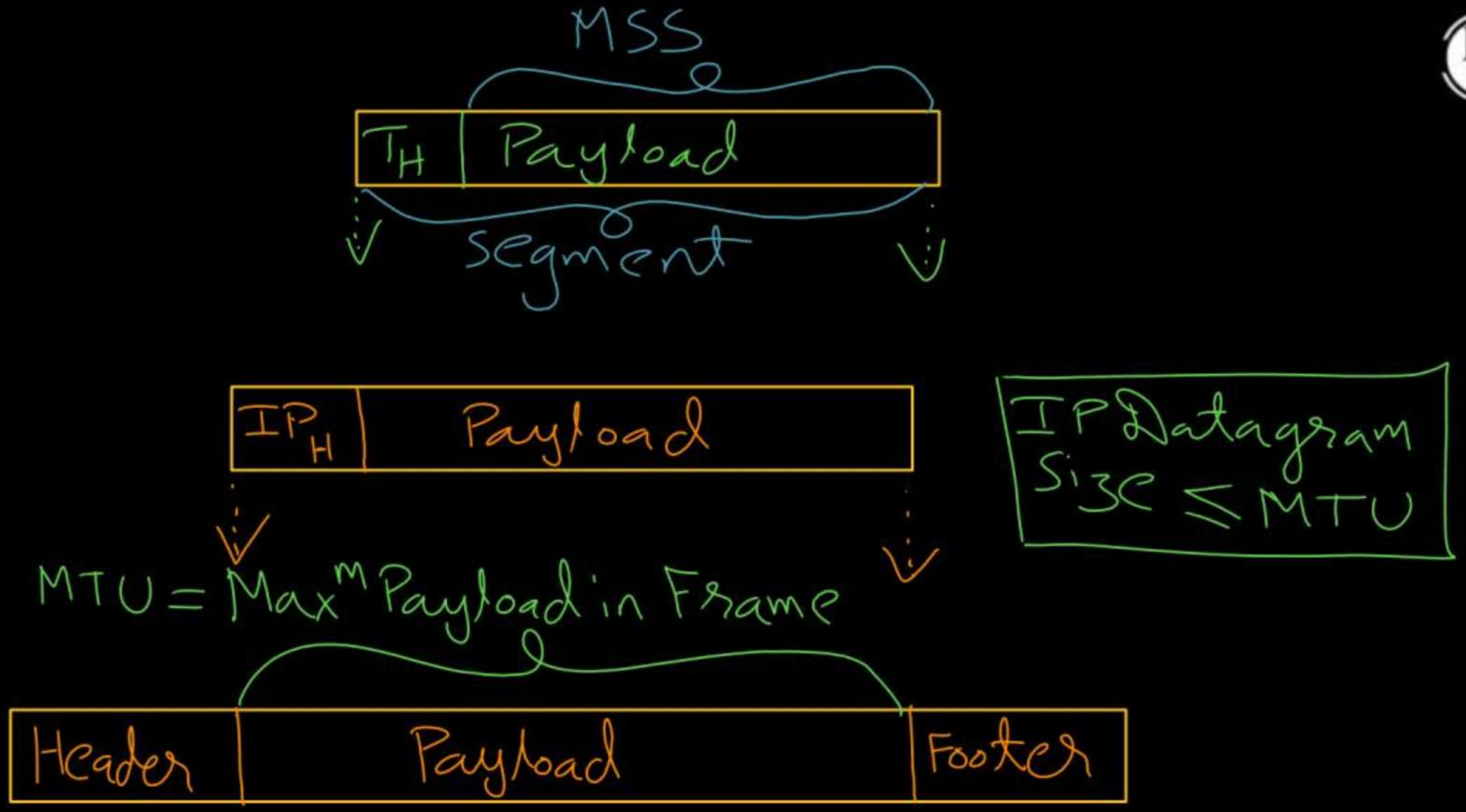
Topic : Maximum Segment Size

→ Maximum Segment Size (MSS)

→ Maximum TCP segment payload size (in bytes)

$$\boxed{\text{MSS} = \text{MTU} - \text{TCP Header Size} - \text{IP Header Size}}$$

→ MSS clamping : MSS is specified during TCP handshake via SYN packet


MSS
T_H
Payload
segment
IP_H
Payload
IP Datagram Size ≤ MTU
MTU = Max^m Payload in Frame
Header
Payload
Footer

TCP Client

Source Network

MTU_1

MSS_1

$MSS = \min(MSS_1, MSS_2)$

TCP Client

TCP Server

SYN

optional Header $= MSS_1$

SYN + ACK

optional Head $= \min(MSS_1, MSS_2)$

R

TCP Server

Dest. Network

MTU_2

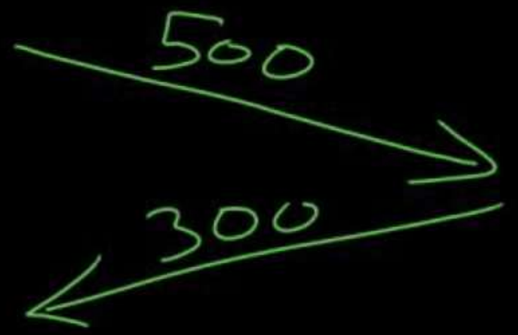
MSS_2

$MSS = \min(MSS_1, MSS_2)$

CASE I

$MSS_1 > MSS_2$

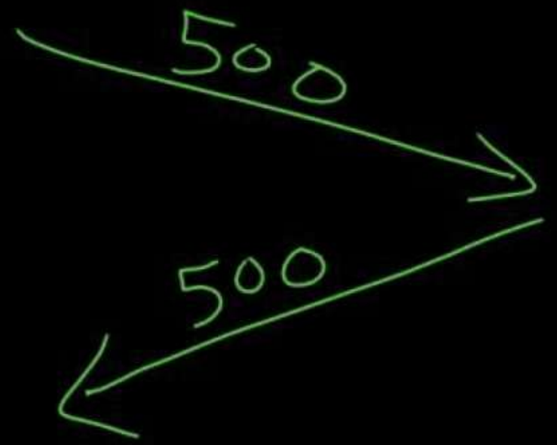
500 300



CASE II

$MSS_1 < MSS_2$

500 700



Topic : Congestion Window

→ Congestion Window Size (CWND)

→ Maintained by TCP sender, to control the congestion

$$[\text{LastByteSent} - \text{LastByteAcked}] \leq \text{minimum}[\text{CWND}, \text{RWND}]$$

→ TCP wants to send a segment contains Y bytes in payload

if $[(\text{LastByteSent} - \text{LastByteAcked}) + Y] > \text{minimum}[\text{CWND}, \text{RWND}]$

→ Not allowed to send

CWND : Sender's Congestion Window Size

RWND : Receiver's advertised available space in Receiving Window

Topic : Congestion Control

→ TCP sender side sequence number space :

1. LastByteAcked
2. LastByteSent

→ Byte numbers [LastByteAcked + 1] to [Last Byte Sent] are sent, but not yet ACKed

$$[\text{LastByteSent} - \text{LastByteAcked}] \leq \text{minimum}[\text{CWND}, \text{RWND}]$$

#Q. On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is _____ .

[GATE-2005]

IIT-B
H.W.

(A) 2048 bytes

(B) 4096 bytes

(C) 6144 bytes

(D) 8192 bytes

Topic : Congestion Control

Algorithm

→ Two phase of TCP congestion control :

1. Slow Start Phase

   [when CWND < SSThresh]

2. Congestion Avoidance Phase

   [when CWND ≥ SSThresh]

SSThresh : Slow Start Threshold

Topic : Slow Start Phase

→ When Congestion Window Size is less then Slow Start Threshold

$\boxed{CWND < SSThresh}$

→ Congestion Window (CWND) grow exponentially

→ CWND increase by 1 MSS on every successful acknowledgment

→ CWND approximately doubles on every round trip time

For every individual ACK
CWND = CWND + 1

#Q. In the slow start phase of the TCP congestion control algorithm, the size of the congestion window :

[GATE-2008]

IISc

(A) does not increase

(B) increases linearly

(C) increases quadratically

(D) increases exponentially

Ans: D

#Q. Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

i. The cwnd increase by 2 MSS on every successful acknowledgement.

ii. The cwnd approximately doubles on every successful acknowledgedment.

iii. The cwnd increase by 1 MSS every round trip time.

iv. The cwnd approximately doubles every round trip time.

Which one of the following is correct?

[GATE-2018, 1-Mark]

IIT-G

H.W.

(A) Only (ii) and (iii) are true

(B) Only (i) and (iii) are true

(C) Only (iv) is true

(D) Only (i) and (iv) are true

Topic : Congestion Avoidance Phase

→ When Congestion Window Size is greater then equal to Slow Start Threshold

$\boxed{CWND \geq SSThresh}$

→ Congestion Window (CWND) grow linearly

→ CWND increase by 1 MSS on every round trip time
[On successful acknowledgments of all transmitted segments]

For every individual ACK

$$CWND = CWND + \frac{1}{\lfloor CWND \rfloor}$$

Topic : Congestion Control

→ When TCP connection established :

Default Initialize : SSThresh = Infinite and CWND = 1 MSS

→ When Time-out occur :

Update : SSThresh = [CWND / 2] and CWND = 1 MSS

$\left\lceil \frac{CWND}{2} \right\rceil$ or $\left\lfloor \frac{CWND}{2} \right\rfloor$

SSThresh = ∞

CWND = 1 MSS → 1

2 MSS → 2

4 MSS → 3

8 MSS → 4

16 MSS → 5

32 MSS → 6

Timeout

Slow Start

SSThresh = $\frac{CWND}{2} = \frac{32\,MSS}{2} = 16\,MSS$

CWND = 1 MSS → 7

2 MSS → 8

4 MSS → 9

8 MSS → 10

16 MSS → 1

17 MSS

18 MSS

19 MSS

Slow Start

Cong. Avoidence

#Q. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____ .

$\boxed{\text{Ans} = 4}$

[GATE-2024, Set-2, 2-Mark]

IISc

CWND = 12 MSS

Timeout

$$SSThresh = \frac{CWND}{2} = \frac{12\,MSS}{2}$$

$$= 6\,MSS$$

CWND = 1 MSS → 1

2 MSS → 2

4 MSS → 3

#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____ .

[GATE-2014, Set-1, 2-Mark]

IIT KGP

H.W.

#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t=0. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time t+60 ms after all acknowledgments are processed is _____

[GATE-2020, 2-Mark]

IIT-D

H.W.

#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

[GATE-2012, 2-Mark]

IIT-D
H.W.

(A) 8 MSS

(B) 14 MSS

(C) 7 MSS

(D) 12 MSS

#Q. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

[GATE-2004]

IIT-D
H.W.

(A) 4000 bytes

(B) 8000 bytes

(C) 10000 bytes

(D) 12000 bytes

2 mins Summary

Topic TCP Congestion Control

Topic [redacted]

THANK - YOU